A finite-step lattice walk in $\mathbb{Z}^3$ with no three collinear vertices

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Abstract

We give an explicit negative answer to Erdős Problem 193. We define a nested infinite discrete Hilbert path $H\colon \mathbb{N} \to \mathbb{N}^2$, select one terminal-steered index from every block of sixteen indices, and lift each selected point by its index. The resulting sequence in $\mathbb{Z}^3$ has successive steps in one fixed finite set and has no three collinear vertices. The obstruction to collinearity is a two-adic pair invariant: for indices with the same terminal quadrant transformation, a valuation of the planar chord equals the ordinary two-adic valuation of the index gap. Applying this identity to the two adjacent chords and their sum rules out a collinear triple. The complete construction has also been formalized in Lean 4 against Mathlib; the formal theorem uses no additional axioms.

1 Introduction

Let $S \subseteq \mathbb{Z}^3$ be finite. An infinite S-walk is a sequence $(P_a)_{a\geq 0}$ in $\mathbb{Z}^3$ such that $P_{a+1} - P_a \in S$ for every a. Erdős Problem 193 asks whether every such walk must contain three collinear points [1, 2]. Gerver and Ramsey proved the corresponding assertion in dimension two and proved a bounded-collinearity result in dimension three [2].

Our answer to the stated three-dimensional question is negative.

Theorem 1.1 (Main theorem)**.** *There are a finite set $S \subseteq \mathbb{Z}^3$ and an infinite sequence $P\colon \mathbb{N} \to \mathbb{Z}^3$ such that*

$$P_{a+1} - P_a \in S \qquad (a \in \mathbb{N}),$$

and no three distinct terms of P are collinear.

The construction uses the discrete Hilbert path, but the space-filling property is not the decisive ingredient. What matters is its finite-state quadrant recursion. After the path H and terminal state σ are defined, the conceptual engine is:

> **Same-state engine.** Put $Q(n) = (H(n), n)$. On every set of indices $E \subseteq \mathbb{N}$ for which σ is constant, the lifted set $\{Q(n) : n \in E\}$ contains no three collinear points.

The proof then has three visibly separate tasks:

1. prove the same-terminal-state pair law for H;
2. construct an infinite bounded-gap set on which σ is constant;
3. prove the abstract same-state engine and apply it to that set.

Section 2 supplies the Hilbert decoder and its finite-state properties. Section 3 proves the pair law, the conceptual core. Section 4 carries out the selector engineering. Section 5 proves and invokes the same-state engine, and Section 6 turns the bounded gaps into a finite step menu. Section 7 is verification and evidence, not an additional premise of the proof.

2 A nested discrete Hilbert path

2.1 A four-state decoder

On a bit pair $(x, y) \in \{0, 1\}^2$, define the four square transformations

$$\begin{aligned} I(x,y) &= (x,y), & S(x,y) &= (y,x), \\ T(x,y) &= (1-y, 1-x), & C(x,y) &= (1-x, 1-y). \end{aligned}$$

The transformations S and T are commuting involutions, with $S \circ T = T \circ S = C$. Here $f \circ g$ denotes function composition: $(f \circ g)(z) = f(g(z))$, so the right-hand transformation acts first. Thus

$$K = \{I, S, T, C\} \cong \mathbb{F}_2^2$$

is closed under composition. These are exactly the states reached by the decoder below.

Let

$$c_0 = (0,0), \quad c_1 = (0,1), \quad c_2 = (1,1), \quad c_3 = (1,0)$$

be the four child positions in the basic U-shaped traversal, and set

$$r_0 = S, \qquad r_1 = I, \qquad r_2 = I, \qquad r_3 = T.$$

For a length-ℓ base-4 word $w = q_{\ell-1} \cdots q_1 q_0$, read the digits from most significant to least significant, starting in state I. If the current state is g, digit q_j emits one pair of coordinate bits and then updates the state by

$$(x_j, y_j) = g(c_{q_j}), \qquad g \longleftarrow g \circ r_{q_j}.$$

The emitted pair at digit position j is used at binary place j:

$$H_\ell(w) = \left(\sum_{j=0}^{\ell-1} x_j 2^j, \ \sum_{j=0}^{\ell-1} y_j 2^j \right). \tag{1}$$

The reading direction and the place values point in opposite directions: the decoder processes $q_{\ell-1}, q_{\ell-2}, \ldots, q_0$ in that order, while q_j writes the coordinate bits at binary place j. Accordingly, for two words "the first mismatch from the low end" means the least $j \geq 0$ for which their digits q_j differ. The state left after the final digit is the terminal state $\sigma(w)$. It is the accumulated quadrant transformation, not a point or an additional coordinate.

2.2 Hilbert properties and terminal states

Lemma 2.1 (Finite-order Hilbert properties)**.** *For every $\ell \geq 0$, the map from length-ℓ base-4 words to $\{0, \ldots, 2^\ell - 1\}^2$ given by (1) is a bijection. Words representing consecutive integers map to lattice neighbors. Moreover, for every length-ℓ word w,*

$$H_{\ell+2}(00w) = H_\ell(w) \quad \textit{and} \quad \sigma(00w) = \sigma(w).$$

Proof. First observe that the decoder respects its current state. Suppose a suffix is decoded from state g instead of I. If the I-started state at some later digit is h, the other state is $g \circ h$: both update on the right by the same r_q. The emitted pair is correspondingly $(g \circ h)(c_q) = g(h(c_q))$. Thus the entire suffix route decoded from g is the bitwise g-transform of the route decoded from I.

Induct on ℓ, recording simultaneously that the order-ℓ route starts at $(0, 0)$ and ends at $(2^\ell - 1, 0)$. The claim is immediate for $\ell = 0$. Put $N = 2^\ell$. At order $\ell + 1$, the first digit places four transformed copies of the order-ℓ route in four disjoint N-by-N quadrants. Bitwise S sends (x, y) to (y, x), while bitwise T sends it to $(N-1-y, N-1-x)$. The four copies therefore have start–end pairs

$$\begin{aligned}(0,0) &\longrightarrow (0, N-1),\\ (0,N) &\longrightarrow (N-1, N),\\ (N,N) &\longrightarrow (2N-1, N),\\ (2N-1, N-1) &\longrightarrow (2N-1, 0).\end{aligned}$$

The maps S and T only swap or reflect square coordinates, so they preserve unit ℓ^1-adjacency inside each transformed copy. The three junction displacements are $(0, 1)$, $(1, 0)$, and $(0, -1)$; they give adjacency between copies. The four disjoint quadrants give bijectivity, and the displayed endpoints complete the simultaneous induction.

Finally, the first leading zero emits $(0, 0)$ and changes the state from I to S. The second emits $S(c_0) = (0, 0)$ and changes the state back to I. The remaining word is therefore decoded in its original state and at its original binary places, proving both padding identities. □

For $n \in \mathbb{N}$, write n in base 4 and prepend zero digits to any sufficiently large even length. Lemma 2.1 makes both parts of the following definition independent of the chosen length.

Definition 2.2. The nested discrete Hilbert path $H : \mathbb{N} \to \mathbb{N}^2$ sends n to the finite-order point of its even-padded base-4 word. Its terminal state $\sigma(n) \in K$ is the transformation left by the same decoding.

To see the infinite claims explicitly, take $m \neq n$ and choose one even length large enough for both padded base-4 words. The words are distinct, so finite-order bijectivity gives distinct points; the padding identity identifies those points with $H(m)$ and $H(n)$. Thus H is injective. Likewise, represent n and $n+1$ at one common even length. Finite-order adjacency, including across a base-4 carry, gives

$$\|H(n+1) - H(n)\|_1 = 1 \qquad (n \in \mathbb{N}). \tag{2}$$

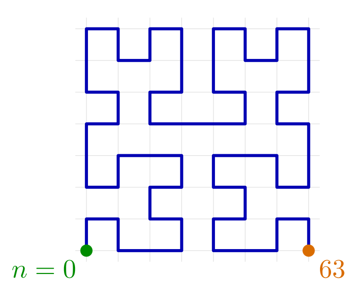

Figure 1: The order-three Hilbert route. Each segment is one planar lattice step; the same four-copy rule produces every higher order.

Only digits 0 and 3 alter the state. Since S and T commute and each undoes itself, for every finite word w,

$$\sigma(w) = S^{N_0(w) \bmod 2} T^{N_3(w) \bmod 2}, \tag{3}$$

where $N_0(w)$ and $N_3(w)$ count its zero and three digits. Thus the terminal state remembers precisely the parities of those two counts.

For later use, every transition $g \mapsto g \circ r_q$ is invertible. If h is the outgoing state after digit q, then the incoming state was $h \circ r_q$, because r_q is an involution. Moreover $r_q(c_q) = c_q$, so the emitted pair can be recovered backwards as

$$(h \circ r_q)(c_q) = h(r_q(c_q)) = h(c_q). \tag{4}$$

This is the mechanism behind the pair law.

3 The two-adic pair law

For a nonzero planar chord u, remove its largest common power of two. The quantity $V(u)$ records twice that binary scale, plus one exactly when both remaining coordinates are odd. Equivalently, it distinguishes the two possible first appearances of the chord: one coordinate becomes nonzero at that scale, or both do.

Formally, put $\nu_2(0) = \infty$. For $u = (u_x, u_y) \in \mathbb{Z}^2 \setminus \{0\}$, define

$$p(u) = \min\{\nu_2(u_x), \nu_2(u_y)\}, \tag{5}$$

$$\epsilon(u) = \begin{cases} 1, & \nu_2(u_x) = \nu_2(u_y), \\ 0, & \nu_2(u_x) \neq \nu_2(u_y), \end{cases} \tag{6}$$

$$V(u) = 2p(u) + \epsilon(u). \tag{7}$$

Lemma 3.1 (Scaling)**.** *For $u \neq 0$ and every positive integer k,*

$$V(ku) = V(u) + 2\nu_2(k).$$

Proof. Multiplication by k adds $\nu_2(k)$ to each finite coordinate valuation. It therefore adds $\nu_2(k)$ to $p(u)$ and leaves the equality bit $\epsilon(u)$ unchanged. □

The factor 2 has a concrete meaning. Multiplication by 2 shifts both coordinate bit streams one place, while V reserves the two consecutive values $2j$ and $2j+1$ for the one-coordinate and two-coordinate appearances at scale j. A shift of one scale must therefore add two to V.

Lemma 3.2 (First mismatch)**.** *Let $q_{\ell-1} \cdots q_0$ and $q'_{\ell-1} \cdots q'_0$ be equally long, even-padded base-4 words with the same terminal state, and let*

$$j = \min\{k \geq 0 : q_k \neq q'_k\}.$$

Thus positions $0, \ldots, j-1$ form their common low suffix. Their Hilbert coordinates agree below binary place j. At place j, exactly one coordinate bit differs when $q_j - q'_j$ is odd, and both coordinate bits differ when $q_j - q'_j \equiv 2 \pmod 4$.

Proof. The words have one common low-digit suffix. Start from their equal terminal states and undo that suffix in lockstep. Invertibility of every transition shows that immediately after the mismatching digit both decoders have one common state h.

Let the mismatching digits be $d \neq e$. Their incoming states were $h \circ r_d$ and $h \circ r_e$. By (4), the emitted pairs at the mismatch are $h(c_d)$ and $h(c_e)$. The corners c_0, c_1, c_2, c_3 form a cyclic Gray code: c_d and c_e differ in exactly one coordinate when $d - e$ is odd, and in both coordinates when $d - e \equiv 2 \pmod 4$. Every $h \in K$ only swaps coordinates and complements coordinate bits. It therefore permutes which coordinates differ but preserves whether the number of differing coordinates is one or two. Starting from the common state h, the identical lower suffix also emits identical bits below place j. □

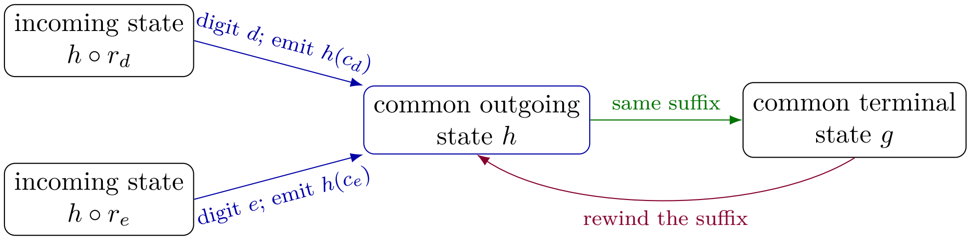

Figure 2: The first-mismatch alignment. Forward decoding sends the two possibly different incoming states through digits d and e to one common outgoing state h. Equal terminal states let the shared low suffix be rewound from g back to h; rewinding changes state, not emitted bits.

Theorem 3.3 (Same-terminal-state pair law)**.** *If $m \neq n$ and $\sigma(m) = \sigma(n)$, then*

$$V(H(n) - H(m)) = \nu_2(|n - m|). \tag{8}$$

Proof. By symmetry assume $m < n$. Pad both base-4 words to one common even length, writing the digits of m as $q_{\ell-1} \cdots q_0$ and those of n as $q'_{\ell-1} \cdots q'_0$. Let j be their first differing position from the low end, and put $d = q_j$ and $e = q'_j$. Then $d \neq e$ and

$$n - m = 4^j M, \qquad M \equiv e - d \pmod 4.$$

If $e - d$ is odd, then $\nu_2(n - m) = 2j$. By Lemma 3.2, exactly one coordinate of $H(n) - H(m)$ has differing bits at position j; in that coordinate the difference has the form

$$\pm 2^j + 2^{j+1} L = 2^j(\pm 1 + 2L)$$

for some integer L. The bracket is odd, so higher binary places cannot cancel the mismatch or add another factor of two. The other coordinate has larger valuation. Thus $p = j$, $\epsilon = 0$, and $V = 2j$.

If $e - d \equiv 2 \pmod 4$, then $\nu_2(n - m) = 2j + 1$. The same odd-bracket calculation applies to both coordinate differences, so both have valuation j. Hence $p = j$, $\epsilon = 1$, and $V = 2j + 1$. These cases exhaust unequal base-4 digits. $\square$

4 A bounded-gap same-state selector

For a block index $a \geq 0$, choose a two-digit suffix according to the state left by a:

state after prefix a	chosen base-4 suffix	$\rho(g)$	full state chain
I	11_4	5	$I \circ I \circ I = I$
S	01_4	1	$I \circ S \circ S = I$
T	31_4	13	$I \circ T \circ T = I$
C	03_4	3	$I \circ C \circ C = I$

The leading I in the last column is the initial decoder state. The middle factor is the net transformation contributed by the prefix a, and the last factor is the net transformation contributed by the chosen suffix when measured separately from I. For example, the S row records the complete path $I \circ S \circ S = I$. Define

$$n_a = 16a + \rho(\sigma(a)). \tag{9}$$

Multiplication by 16 appends two base-4 digits. If the prefix contributes $g = \sigma(a)$, the chosen suffix contributes a second copy of g. Starting from I, the completed word therefore has the full state chain

$$\sigma(n_a) = I \circ g \circ g = I, \tag{10}$$

because every element of K is its own inverse. Hence all selected indices have one common terminal state.

There is one selected index in every block of sixteen. Since $\rho(K) = \{1, 3, 5, 13\}$,

$$\begin{gathered} n_{a+1} - n_a = 16 + \rho(\sigma(a+1)) - \rho(\sigma(a)), \\ 4 \le n_{a+1} - n_a \le 28. \end{gathered} \tag{11}$$

In particular, (n_a) is strictly increasing and infinite. Lift the selected planar points by using the selected index as height:

$$P_a = (H(n_a), n_a) \in \mathbb{Z}^3. \tag{12}$$

The bounded gaps will give a finite step menu in Section 6. This selector is an engineering step: bounded gaps are needed for the finite menu, while the no-collinearity argument below uses only the shared terminal state.

5 Excluding collinear triples

Theorem 5.1 (Same-state lift theorem)**.** *Let $E \subseteq \mathbb{N}$ and suppose that σ is constant on E. Put $Q(n) = (H(n), n)$. Then $\{Q(n) : n \in E\}$ contains no three collinear points.*

Proof. Suppose for contradiction that $a < b < c$ are elements of E for which $Q(a), Q(b), Q(c)$ are collinear. Their terminal states agree by hypothesis, so the pair law applies to all three pairs. Write the adjacent height gaps and planar chords as

$$\begin{aligned} A &= b - a, & B &= c - b, \\ X &= H(b) - H(a), & Y &= H(c) - H(b). \end{aligned}$$

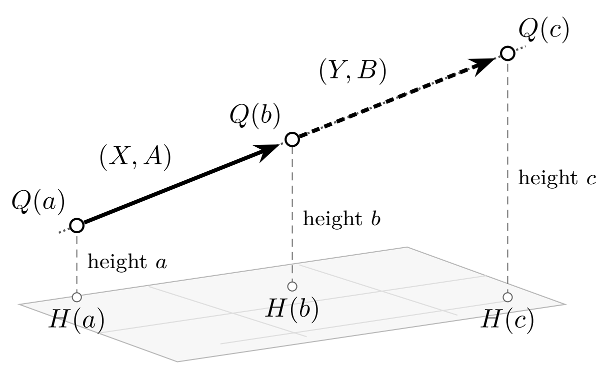

Figure 3: The contradiction setup, with $Q(n) = (H(n), n)$. The solid arrow labeled (X, A) and the dashed arrow labeled (Y, B) are the directed adjacent displacements. The dotted guide through $Q(a), Q(b), Q(c)$ marks the temporary collinearity assumption.

The full displacements are (X, A) and (Y, B). The three lifted points are collinear over $\mathbb{R}$ exactly when these displacements are proportional. Because their last coordinates satisfy $A, B >$

0, the proportionality factor is forced to be B/A. Clearing denominators gives the equivalent coordinatewise equation

$$BX = AY. \tag{13}$$

Write

$$A = gr, \qquad B = gs, \qquad g = \gcd(A, B), \qquad \gcd(r, s) = 1.$$

After cancelling g, equation (13) becomes $sX = rY$. For each coordinate i, the relation $sX_i = rY_i$ and $\gcd(r, s) = 1$ imply $r \mid X_i$. Hence $W = X/r$ is an integer vector. Substitution into $sX = rY$ then gives $Y = sW$, so

$$X = rW, \qquad Y = sW. \tag{14}$$

Injectivity of H gives $X \neq 0$, hence $W \neq 0$.

Apply the pair law and scaling lemma first to (a, b) and then to (b, c):

$$\begin{aligned}\nu_2(g) + \nu_2(r) &= V(X) = V(W) + 2\nu_2(r),\\ \nu_2(g) + \nu_2(s) &= V(Y) = V(W) + 2\nu_2(s).\end{aligned}$$

Hence $\nu_2(r) = \nu_2(s)$. Since r and s are coprime, both valuations are zero. Thus r and s are odd, and the same equations give

$$V(W) = \nu_2(g).$$

Now use the endpoint pair (a, c). Its index gap and planar chord are

$$c - a = g(r + s), \qquad H(c) - H(a) = (r + s)W.$$

Put $t = \nu_2(r + s)$. The pair law and scaling lemma give

$$\nu_2(g) + t = V((r + s)W) = V(W) + 2t = \nu_2(g) + 2t.$$

Therefore $t = 0$, so the endpoint pair forces $r + s$ to be odd. But the two adjacent pairs forced both r and s to be odd, whose sum is even. This is a contradiction. □

Corollary 5.2 (The selected lift is triple-free)**.** *The sequence* $(P_a)_{a\geq 0}$ *defined by* (12) *contains no three collinear terms.*

Proof. The indices n_a are strictly increasing, satisfy $\sigma(n_a) = I$, and obey $P_a = Q(n_a)$. Apply Theorem 5.1 to $E = \{n_a : a \in \mathbb{N}\}$. □

The use of all three pair laws—for gaps A, B, and $A + B$—is essential. A pairwise direction restriction alone would not rule out every triple.

6 The finite step menu

The third-coordinate difference of one successive step is

$$d_z = n_{a+1} - n_a, \qquad 4 \leq d_z \leq 28.$$

The Hilbert path takes one planar lattice step per unit increase of its index. Therefore (2) and the triangle inequality give the sharper path-length bound

$$|d_x| + |d_y| = \|H(n_{a+1}) - H(n_a)\|_1 \leq n_{a+1} - n_a = d_z.$$

Hence every successive displacement belongs to the fixed finite set

$$S = \{(d_x, d_y, d_z) \in \mathbb{Z}^3 : 4 \le d_z \le 28,\ |d_x| + |d_y| \le d_z\}. \tag{15}$$

The strictly increasing third coordinate makes (P_a) infinite with no repeated terms, and Corollary 5.2 excludes three collinear terms. Equations (12) and (15) therefore prove Theorem 1.1.

Remark 6.1. The displayed set S is a convenient finite superset of the realized step vectors, not a minimal menu. The independently reconstructed first 500,000 steps realize 16 distinct vectors.

7 Verification and evidence

Proof boundary. The proof of Theorem 1.1 ends above. Nothing in this section is an additional premise; it records formal verification, implementation checks, and the status of external claims.

7.1 Lean formalization

The repository [6] contains a Lean 4 formalization in `formal/Hilbert193`. It uses Lean 4.33.0 and the corresponding pinned Mathlib revision. The final theorem is `Hilbert193.erdos193_unconditional` in `Hilbert193/Continuity.lean`. It constructs:

1. a finite set of integer three-dimensional step vectors;
2. an explicit sequence $\mathbb{N} \to \mathbb{Z}^3$ whose successive displacements lie in that set; and
3. a proof that every ordered triple of distinct indices fails the exact integer collinearity equations.

The Lean predicate packages collinearity as the two integer cross-multiplication equalities forced by the strictly increasing third coordinate. As shown in the proof above, this is the denominator-free form of ordinary real affine collinearity for the ordered points in this walk.

The axiom audit in `Hilbert193/AxiomAudit.lean` reports only `propext`, `Classical.choice`, and `Quot.sound`, the standard axioms used by Mathlib. No theorem in the construction depends on a finite computation. From the repository root, the formal proof is checked by

```
cd formal/Hilbert193
lake build
lake env lean Hilbert193/AxiomAudit.lean
```

7.2 Finite implementation checks

The independent finite reconstruction is checked by

```
python3 verify_hilbert_construction.py hilbert-193-500k.jsonl \
  --steps 500000 --fresh
```

It reconstructs 500,001 vertices and checks the defining selector, coordinates, and fixed-menu condition for the recorded finite artifact. It does not enumerate triples. A separate exact direction-hash scan is run by

```
clang++ -O3 -std=c++17 -pthread verify_hilbert_exhaustive.cpp \
  -o verify_hilbert_exhaustive
./verify_hilbert_exhaustive hilbert-193-500k.jsonl
```

For each vertex, this scan reduces the vectors to all earlier vertices to canonical primitive integer directions. A repeated direction is equivalent to a collinear triple. The completed four-worker run checked all 125,000,250,000 pairs in 6299.67 seconds and found no repetition. The artifact SHA-256 is

```
6f8fdf59b7ecef5533b8a65265d80df9
61f67a955dabaf9d5bb973d7ae143d63.
```

Concatenating the two lines gives the digest.

7.3 Logical and historical status

The theorem above is unconditional. The Lean development checks the formal statement; the finite reconstruction and exhaustive scan check an implementation and a finite prefix only. Neither computation proves the infinite theorem. Novelty, priority, and correspondence with the intended reading of Erdős Problem 193 remain subject to external mathematical review and community acceptance.

Acknowledgments and disclosure

The working stack included Paseo, Oh My Pi, and OpenAI GPT-5.6 for proof exploration, Lean code drafting, computational verification tooling, visualization, and manuscript preparation. The author directed the work, checked the generated artifacts, and takes responsibility for every claim. The proof is kernel-checked, but at the date of this preprint it has not yet completed external mathematical peer review.

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